Semester V												
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
MOOC	Open Elective 2 (NPTEL/SWAYAM/MOOC)	3	0	0	3	3	25	25	50	--	--	100
24HS303T	Petroleum Engineering Economics	3	0	0	3	3	25	25	50	--	--	100
24PE331T	Health Safety and Environment	3	0	0	3	3	25	25	50	--	--	100
24PE332T	Petroleum Refinery Engineering											
24PE301T	Well completion and fundamentals of Production Engineering	2	1	0	3	3	25	25	50	--	--	100
24PE302T	Well Test Analysis	2	1	0	3	3	25	25	50	--	--	100
24PE303T	Advanced Drilling Technology	3	1	0	4	4	25	25	50	--	--	100
24PE304P	Introduction to Petroleum Software Laboratory	0	0	2	2	1	--	--	--	50	50	100
24PE305P	Petroleum Engineering –I Laboratory	0	0	2	2	1	--	--	--	50	50	100
24INT251	Industrial Orientation	0	0	0	0	0	--	--	--	--	--	--
	Total	**15**	**3**	**4**	**23**	**21**						

Semester VI												
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
	Open Elective 3(From Other Department of FoET)	3	0	0	3	3	25	25	50	--	--	100
24PE333T	Data Analytics for Petroleum Engineers	2	0	0	2	2	25	25	50	--	--	100
24PE334T	Carbon Utilization and Storage											
25PE331T	Natural Gas Engineering and LNG Value Chain	3	0	0	3	3	25	25	50	--	--	100
24PE336T	Pipeline Engineering											
24PE306T	Reservoir Modelling and Simulation	2	1	0	3	3	25	25	50	--	--	100
24PE307T	Surface and Offshore Production Operations	2	1	0	3	3	25	25	50	--	--	100
24PE308T	Unconventional Hydrocarbon Energy Resources	3	0	0	3	3	25	25	50	--	--	100
24PE309P	Petroleum Engineering – II Laboratory	0	0	2	2	1	--	--	--	50	50	100
24PE310P	Petroleum Product Testing Laboratory	0	0	2	2	1	--	--	--	50	50	100
	Total	**15**	**2**	**4**	**21**	**19**						

24HS303T					Petroleum Engineering Economics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To provide students with basic economic concepts for decision-making in various petroleum engineering projects.
- To apply these concepts to the real world using applied examples and case studies.
- To develop the fundamental understanding of economics and costing for engineering operations.
- To acquire knowledge of economics to facilitate the process of economic decision-making in non-renewable markets.

Unit I: Overview of Petroleum Engineering Economics **Hours: 08**

Meaning and scope of petroleum engineering economics, fundamentals of economics: scarcity and choice; renewable vs non-renewable, role of petroleum sector in development, objective of a firm, production possibility curve; types of efficiency

Unit II: Demand and Supply **Hours: 10**

Law of demand and exceptions to the law; demand schedule, demand curve; Law of supply and exceptions to the law; supply schedule; supply curve; movements vs shifts; elasticity of demand and supply, Game theory

Unit III: Theory of Production and Cost **Hours: 12**

Production function and Types; law of variable proportions; Traditional and Modern theory of cost; types of cost; relationship between cost curves and product curves; break-even analysis

Unit IV: Markets for non-renewable energy **Hours: 09**

Competition in fossil fuel and gas industry, Issues of energy security, Energy price regulations and mechanism; Supply chain (Case studies)

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Define the fundamental concepts of petroleum engineering economics
CO2- Describe various cost concepts and maintenance of break-even scenario for a business.
CO3- Analyze the demand and supply problems of the firm using key concepts.
CO4- Demonstrate the applicability of concepts like game theory, energy price regulations and competition in non-renewable markets.
CO5 - Evaluate the production and supply chain of the firm in order to improve economic efficiency.
CO6- Design and analyze business models using case studies.

TEXT / REFERENCE BOOKS

1. Riggs, J. L., Bedworth, D. D., & Randhawa, S. U. (2019). *Engineering economics* (8th ed.). McGraw-Hill Education.
2. Mankiw, N. G. (2018). *Principles of economics* (8th ed.). Cengage Learning.
3. Schwarz, P. M. (2013). *Energy economics*. Routledge.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE331T					Health Safety and Environment					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To understand various hazards associated with the oil and gas industry.
- Discuss various case studies on hazards and risk analysis.
- To study and demonstrate the requirements for safety.
- To develop the skills to address the Safety issue.

Unit I: Introduction **Hours: 10**

Introduction to Physical, Chemical, Biological, Environmental, Ergonomical etc. Hazards and their remediations. Environmental issues and management, Environmental impact and management Impact of oil and gas industry in the marine environment, Oil hydrocarbons in marine environment, Chemical disposal of the offshore industry and environmental management, Dispersion models and atmospheric pollution Dispersion models continued, Hazard assessment

Unit II: Hazard And Risk Analysis **Hours: 10**

Operational Safety, Introduction to HSE Safety assurance, Safety in design and operations, Organizing for safety Hazard, classification and assessment, Hazard evaluation and control, Hazop, FMEA, Risk Matrix, HAZID, HAZOP, QRA (API 14 J, OISD), Safe Work Practices: PTW, MOC, SIMOPS etc. (ref API RP 75, OISD, OMR), Electrical Safety; Classification of Hazardous locations, Hazardous area (Ref IER, OISD, OMR, API RP 500 & 14 F) Accident Investigations: Study of major Accidents like Piper Alpha, Flixborough, Bhopal etc., Investigation techniques Emergency Response planning Audits & Inspection. Audit methodology, protocol, typical checklists for Drilling rigs, Work over activities, logging, etc.

Unit III: HSE Management System **Hours: 10**

Accident modeling, risk assessment & management, Dose assessment, safety regulations, Toxic releases- models and methods Chemical risk analysis Quantitative risk assessment, Fire Risk and Control, Flammability diagrams, Fire and explosion: prevention methods, Event tree and fault tree analyses, Process safety management, Software used in HSE, OISD, API RP 75, ISO, OSHAS Standards, OMR and Petroleum Rules.

Unit IV: Environmental Pollution and Control **Hours: 09**

Environment Concepts: Effect on eco-system; Air, Water, & Soil of HCs, Environmental Impact Assessment, Oil Spills Control and their Management. Maritime Environmental Rules & Regulations. Drilling / Oil Storage / Effluent water / waste (solid & sludge) treatments their disposal and remediation of soil etc. Directorate of Mine Safety, DGMS. Downstream Safety: Implementing Agency PNGRB; Safety Regulations (Technical Standard, Specification and Safety Standards T4S), Emergencies, Mutual Aida; Emergency Response and Disaster Management Plan ERDMP)

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Understand the various hazard associated with oil and gas industry
CO2- Apply the control methods & safety measures for preventing hazards and risk
CO3- Analyse the various risk matrixes, safe work practices and accident investigations
CO4- Implement the standards of HSE management
CO5 - Evaluate the various environmental issues in oil and gas industry and safety regulations
CO6- Demonstrate the various safety issues during drilling and refinery operations

TEXT / REFERENCE BOOKS

1. Lees, F. P. (1996). *Lees' loss prevention in the process industries: Hazard identification, assessment, and control* (2nd ed.). Butterworth-Heinemann.
2. Peavy, H. S., Rowe, D. R., & Tchobanoglous, G. (1985). *Environmental engineering*. McGraw-Hill.
3. Sanders, R. E. (2011). *Chemical process safety: Learning from case histories* (4th ed.). Butterworth-Heinemann.
4. National Fire Protection Association. (2024). *NFPA codes and standards*. National Fire Protection Association.
5. Marshall, V. C., & Ruhemann, S. (1998). *Fundamentals of process safety*. Institution of Chemical Engineers.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE332T					Petroleum Refinery Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To provide the concept of petroleum refining and explain the different methods of petrochemical reactions and their applications
- To provide the importance of various refining processes and their applications
- To explain the significance petrochemicals productions

Unit I: Petroleum Crude and Products Characterization **Hours: 10**

Types of crudes, crude composition Characteristics and classification–Crude oil properties. IS 1448: Standard –Testing of Petroleum crude Products: Specifications and their Significance.

Unit II: Processing of Petroleum and Treatment Techniques **Hours: 10**

Pre-treatment of crude for Refining–Dehydration and desalting–Atmospheric distillation, Vacuum distillation of residue products–Treatment techniques for vacuum distillates with different processes like solvent extraction –DE asphalting, dewaxing, hydro fining, catalytic dewaxing and clay contact process–Production of lubricating oils. Hydro cracking, principles, process requirements, product yields and qualities and reside-cracking –Hydrotreating –Sulphur removal, hydro finishing.

Unit III: Thermal and Catalytic Cracking **Hours: 10**

Thermal cracking – Processes, operating parameters, feed stock selection and product yields, Advantages –Types and functions of secondary processing – Visbreaking – Processes, operating parameters and advantages–Coking –Operating parameters and advantages. Fluid catalytic cracking –processes, operating parameters, feed stock selection and product yields –Advantages.

Unit IV: Upgradation of Refining Products **Hours: 09**

Principle, Processes, Operating Parameter and advantages of Reforming – Isomerisation – Alkylation – Polymerization. Asphalt manufacture, Air blowing technology, Bitumen Types and their properties, Acid gas removal and sulphur removal techniques.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Introductory information about origin, exploration and production of petroleum crude and understand their properties with the help of standard testing protocols

CO2- Recognize various primary crude processing techniques like distillation, de-asphalting, hydrocracking, hydrotreating, and their supporting processes.

CO3- Recognize various secondary thermal cracking, catalytic cracking and reforming and coking and their supporting processes.

CO4- Understanding the process technologies for reforming, isomerization, alkylation and polymerization unit process.

CO5 -To understand and evaluate various residue processing schemes.

CO6- To apply the finishing processes to petroleum products for meeting the market specifications in view of fuel quality and environmental regulations.

TEXT / REFERENCE BOOKS

1. Rao, B. K. B. (1984). *Modern petroleum refining processes* (5th ed.). Oxford & IBH Publishing.
2. Rao, B. K. B. (2004). *A text book on petrochemicals* (5th ed.). Khanna Publishers.
3. Dryden, C. E., Rao, M. G. (Ed.), & Sittig, M. (Ed.). (1973). *Outlines of chemical technology* (2nd ed.). Affiliated East-West Press.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

PART A: 10 Questions of 2 marks each-No choice **20 Marks**

PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE301T					Well Completion and Fundamentals of Production Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	1	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To provide an Introduction to Petroleum Production Operations with an emphasis on the basic knowledge of petroleum production systems and operations.
- To provide the concepts of different well problems and well intervention methods used.

Unit I: Introduction To Petroleum Production System **Hours: 12**

Petroleum Production System, Equipment of naturally flowing well : Christmas tree, valves, hangers, flow control devices, packers, tubular and flow lines. Surface and subsurface chokes. Fundamentals of Well Testing, Well Completion: Well-completion Methods, Perforating Oil & Gas Wells - Conventional and Unconventional techniques viz. through tubing and tubing conveyed underbalanced perforating techniques, type size and orientation of perforation holes. Smart wells- intelligent completions.Bean size selection, Principle and application of workover methods/ReCompletion, Workover fluids,- Well Perforations instruments and techniques, Connecting the payzone and bore hole: Well activation methods, use of compressed air & liquid Nitrogen.

Unit II: Components of Production System **Hours: 09**

Reservoir performance, Supplementing reservoir energy: Fluid Injection and Artificial lift methods, Introduction to inflow performance, Productivity index. Formation damage diagnosis, Skin effect, IPR in case of different drive mechanism. Vogel IPR equation. Composite production system: Completion pressure drop, Pressure drop in tubing, multiphase flow regimes. Choke performance, types of chokes. Overall production system pressure losses.

Unit III: Well Productivity Evaluation **Hours: 09**

Production logging and its application, Problems during logging in deviated and horizontal wells, Spinner Flow meters and spinner crossplots, Fluid identification tools (Temperature log, Nuclear fluid density, GHOST, and DEFT) Advanced logging tools (Flow Scan Imager Tool and Multiple Array Production Suit, and Noise logs).

Unit IV: Production System Analysis and Optimization **Hours: 09**

Production System Analysis: PI & IPR of self-flowing wells. Single and Multiphase flow in tubing and flow-lines. Sizing, selection and performance of Tubing, chokes and surface flow lines. Production Optimization – Nodal System analysis.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Explain the fundamentals of petroleum production systems, well completion methods, and production equipment.
CO2- Apply well completion, perforation, and workover techniques to improve well productivity.
CO3- Analyze reservoir and production system performance using IPR, productivity index, and pressure loss concepts.
CO4- Evaluate the performance of tubing, chokes, and multiphase flow in production systems.
CO5 -Interpret production logging data and advanced logging tools for well performance evaluation.
CO6-Apply nodal analysis and production optimization techniques for efficient well production.

TEXT / REFERENCE BOOKS

1. Guo, B. (2007). *Computer-aided petroleum production engineering*. Elsevier.
2. Gatlin, C. (1960). *Petroleum engineering: Drilling and well completion* (1st ed.). Prentice Hall.
3. Nind, T. E. W. (1981). *Principles of oil well production* (2nd ed.). McGraw-Hill.
4. Allen, T. O., & Roberts, A. P. (1982). *Production operations* (Vol. 1: Well completions, workover, and stimulation). Oil & Gas Consultants International.
5. Allen, T. O., & Roberts, A. P. (1984). *Production operations* (Vol. 2: Equipment and tools). Oil & Gas Consultants International.
6. Schlumberger. (1987). *Fundamentals of production logging*. Schlumberger Educational Services.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE302T					Well Test Analysis					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	1	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To Understand the various transient well tests conducted during the various stages of life of a producing field.
- To have a broad knowledge regarding the well-test data acquisition.
- To Interpret the Hydrocarbon properties therein and Reservoir characteristics from the test results.

Unit I: Introduction to the Well Test Analysis **Hours: 06**

Purpose and objectives of well testing, Types of well tests: drawdown, buildup, injectivity, falloff, multi-rate, Basic reservoir flow models: radial, linear, boundary-dominated, Data acquisition and quality control

Unit II: Pressure Transient Analysis for Single-Phase Flow **Hours: 13**

Concept of pressure diffusivity, Analysis of drawdown and buildup tests in infinite-acting radial flow, Type curve analysis, Effects of skin and wellbore storage, Specialized plots: Horner, MDH, derivative analysis, Introduction to well-test analysis software

Unit III: Well Test Analysis in Complex Systems **Hours: 10**

Partially completed and limited entry wells, naturally fractured reservoirs (dual porosity models), Layered reservoirs, Horizontal wells, Interference testing

Unit IV: Production System Analysis and Optimization **Hours: 10**

Well testing in gas reservoirs (pseudo pressure, pseudo time), Multiphase Flow Considerations, Integration of well test data with other reservoir data (seismic, well logs, production data), Case Studies and real-world examples.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Identify the principle governing equations of transient well testing and their solutions.
CO2- Understand the various transient well-testing methods and their requirement for petroleum exploitation activities.
CO3- Apply the methods to interpret well test data to determine petrophysical parameters in dynamic flow conditions around a well.
CO4- Classify the standard procedure to control the well.
CO5 -Appraise the Petrophysical properties from the well-bore vicinity to the extent of the reservoir.
CO6-Formulate an appropriate well-testing program for the formation.

TEXT / REFERENCE BOOKS

1. Bourdet, D. (2002). *Well test analysis: The use of advanced interpretation models*. Elsevier.
2. Lee, J. (1982). *Well testing*. Society of Petroleum Engineers.
3. Earlougher, R. C., Jr. (1977). *Advances in well test analysis*. Society of Petroleum Engineers of AIME.
4. Dake, L. P. (1978). *Fundamentals of reservoir engineering*. Elsevier.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

PART A: 10 Questions of 2 marks each-No choice **20 Marks**

PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE303T					Advanced Drilling Technology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	-	-	100

COURSE OBJECTIVES

- Understanding of directional and horizontal drilling, well profile monitoring of the progressing well path and corrective measures for deflected well trajectory.
- Comprehensive knowledge on the emerging trends in drilling technology.
- Comprehend the well-control methods and its procedures.

Unit I: Directional Drilling **Hours: 15**

Introduction to directional drilling technology; Objectives of directional drilling; Directional well profiles and well path calculations; Tools for deflection & orientation; Downhole mud motors: Dynadrill and Turbodrill; Power calculations and applications of mud motors; Directional drilling problems & their remedies; Well surveying objectives, & tools for surveying/methods; Well path correction methods and its execution in drilling.

Unit II: HORIZONTAL WELL DRILLING **Hours: 15**

Introduction to horizontal well drilling; Horizontal well drilling objectives; Horizontal drilling techniques; Horizontal well profiles; Extended reach drilling, Special mud requirements and their characteristics for horizontal drilling; Objectives of MWD / LWD tools; Telemetry system and data interpretation; Geo steering: Auto and Verti-track systems; Surveying analysis and calculations for well coordinates; Horizontal drilling problems and their remedies.

Unit III: EMERGING TRENDS IN DRILLING **Hours: 11**

Coil tubing drilling technique and application, Slant drilling techniques and the specialized tools, Introduction to multilateral drilling, Technology advancement of multilaterals, Slim hole drilling, Casing While Drilling, Prospects of big data analytics in drilling: Automation and robotics in drilling operations

Unit IV: Well Control Principles & Procedures **Hours: 11**

The anatomy of a well kick and blowouts: Problem, symptoms and controlling measures; Kick control methods: Dynamic kick control, Driller method & Engineer methods, Well killing calculations.

Max <52Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Demonstrate the different methods of directional drilling.
CO2- Apply the fundamentals of Horizontal well drilling techniques
CO3- Execute surveying tools and MWD/LWD for monitoring the bottom hole condition
CO4- Analyse the well profile and survey the well trajectory
CO5 -Defend the requirement for the implementation of emerging technologies in the conventional directional drilling process.
CO6- Investigate and execute well killing operation.

TEXT / REFERENCE BOOKS

1. Inglis, T. A. (1987). *Petroleum engineering and development studies* (Vol. 2: *Directional drilling*). Graham & Trotman.
2. Rabia, H. (1986). *Oil well drilling engineering: Principles and practices*. Graham & Trotman.
3. Bourgoyne, A. T., Jr., Chenevert, M. E., Millheim, K. K., & Young, F. S., Jr. (1991). *Applied drilling engineering*. Society of Petroleum Engineers.
4. Joshi, S. D. (1991). *Horizontal well technology*. PennWell Publishing Company.
5. Azar, J. J., & Samuel, G. R. (2007). *Drilling engineering*. PennWell Books.
6. International Association of Drilling Contractors. (2000). *IADC drilling manual*. International Association of Drilling Contractors.
7. Adams, N. J. (1980). *Well control: Problems and solutions*. Petroleum Publishing Company.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE304P					Introduction to Petroleum Software Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To provide an overview of the oil and gas industry software tools.
- To provide the importance of modelling and simulations for different petroleum engineering problems.
- To explain how to identify the best tool matching the type and scope of numerical study.

LIST OF EXPERIMENTS

1. Analytical Modelling for Buckley Leverett Solution using any programming language-Part A
2. Analytical Modelling for Buckley Leverett Solution using any programming language-Part B
3. Developing a Numerical Model for One Dimensional Flow through Porous Media Problem – Part A
4. Developing a Numerical Model for One Dimensional Flow through Porous Media Problem – Part B
5. Reservoir Modelling and Simulation Using IMEX/GEM/tNavigator – Part A
6. Reservoir Modelling and Simulation Using IMEX/GEM/tNavigator – Part B
7. History Matching Problem – Part A
8. History Matching Problem – Part B
9. Integrated Static and Dynamic Modeling using tNavigator.
10. Introduction to Well Testing using Saphire/Topaze/ Emeraude.
11. Design of Hydraulic fractures for Mini-frac and fracture applications using FracPro – Part A
12. Design of Hydraulic fractures for Mini-frac and fracture applications using FracPro – Part B
13. Introduction to Integrated Production Modeling Suite – Part A
14. Introduction to Integrated Production Modeling Suite – Part B

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Demonstrate the role of simulation software in the engineering industry and in specific to the upstream petroleum industry

CO2 - Classify the various software tools available in the individual domain of upstream petroleum industry like seismic data processing and interpretation, reservoir modelling and simulation, drilling, and production engineering.

CO3 - Build a reservoir simulation model using CMG (reservoir simulation software) and simulate the specific initial and boundary constraints defined.

CO4 -Excel the fundamental modelling workflows associated with the simulation software like conceptualizing, mathematical modelling, numerical modelling and solving the set of equations using the gauss elimination technique.

CO5 -Comprehend complex and dynamic nature of the petroleum engineering problems and formulate a solution strategy for effective management at the field scale.

CO6 -Identify the best tool matching the type and scope of the numerical study the student has been deployed to perform in the future.

TEXT / REFERENCE BOOKS

1. Peaceman, D. W. (1977). *Fundamentals of numerical reservoir simulation*. Elsevier.
2. Fanchi, J. R. (2006). *Principles of applied reservoir simulation* (3rd ed.). Gulf Professional Publishing.
3. Pinder, G. F., & Gray, W. G. (2008). *Essentials of multiphase flow and transport in porous media*. John Wiley & Sons.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**

End semester examination and Viva-voce **50 Marks**

24PE305P					Petroleum Engineering – I Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To provide basic knowledge related to petroleum engineering laboratory equipment.
- To teach the fundamental aspects of petroleum engineering, which includes core plug making, core trimming and swabbing.
- To determination of permeability and porosity of the core sample by different methods and compared with each other for the research point of view.

LIST OF EXPERIMENTS

1. To make core plug ready for experiment in Core Plugging.
2. To perform the Core Trimming and Swabbing
3. To determine the Effective porosity of given core sample by saturation method.
4. To determine the Porosity of given core sample by Helium Porosimeter.
5. To determine the Permeability of given sample by using Ruska Liquid Permeameter.
6. To determine the Permeability of the core sample by using Gas Permeameter.
7. To study the core flooding apparatus under reservoir pressure and temperature conditions.
8. To study the petrophysical properties of core in core flooding apparatus.
9. To determine the viscosity of oil by using capillary viscometer.
10. To study of the various Drilling Equipment's.
11. To study of the Sucker Rod Pumping System: A Case Study
12. To study of the working of Gas Lift system.
13. Study and understanding of Group Gathering Sation (GGS) system.

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand and demonstrate core preparation techniques for various core flooding environments
CO2 - Determination of the petrophysical properties.
CO3 - Design of enhanced oil recovery experiment using core flooding apparatus.
CO4 -Determination of viscosity of light oil samples.
CO5 -Understand the drilling equipment and production process.
CO6 -.Demonstration of artificial lift equipment.

TEXT / REFERENCE BOOKS

1. Dandekar, A. Y. (2013). *Petroleum reservoir rock and fluid properties* (2nd ed.). CRC Press.
2. Begg, H. D. (1998). *Production optimization*. Oil & Gas Consultants International (OGCI) Publications.
3. Dake, L. P. (1978). *Fundamentals of reservoir engineering*. Elsevier.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**

End semester examination and Viva-voce **50 Marks**

24INT251					Industrial Orientation					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	0	0	0	-	-	-	50	50	100

COURSE OBJECTIVES

- The objective of the course is to orient students towards entire spectrum of the Oil and Gas Industry best working practices.

Industrial Orientation

The students are required to visit the entire value chain of the Oil and Gas industry for

a) Acquaintance with the best working practices adopted in the industry.
b) Understanding the industry working environment.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Acquire Technical skills prevailing in the industry.
CO2 - Develop professional communication skills (both written and oral).
CO3 - Demonstrate capability for understanding project plan and project management technique
CO4 - Demonstrate capability for managing team and working as team leadership.
CO5 -.Develop Professional vision to chart its own professional life.
CO6 -.Demonstration of artificial lift equipDevelop skill sets of professional and ethical petroleum engineering practices

END SEMESTER EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**
End semester examination and Viva-voce **50 Marks**